

Articulated Automaton

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Introduction

The impact of nonsmooth nonlinearities, specifically dead zones and singularities, within actuators on motion precision have presented a notable challenge in control research, often disregarded due to their time-variant characteristics. However, recognizing and addressing these nonlinearities enables pre-compensation strategies, enhancing the effectiveness of motion through dead zones. By accounting for these complexities, advancements in motion control can be achieved, mitigating imprecisions and improving overall performance in various applications.

MuJuCo Simulations

To use the MuJuCo Simulations we opted for the use of a python script with a MuJuCo binding. Using the MuJuCo software itself did not allow full control over all of the parameters and variables. We tried to use the google colab which was a tutorial demonstrating the use of MuJuCo in python, but had limited GPU allowance.

To find out deadzones of the shadowhand we had to use unorthodox methods to derive physical values. We used the control variable, which had an index value for each link on the hand. This was synchronous with the control variables used in the MuJuCo software itself, but could be used in looping python code. To find all the reachable zones we passed each link through the range of minimum and maximum values each link could have in a nested for loop, having the end effector in the innermost loop and moving outwards from there. Through this, every reachable zone would be mapped out using the `data.xpos[id of link]` value at each iteration of the loop. We had to use a 5 degree increment since less than that would create data files that were too much to handle.

The deadzone is intuitively the inverse of the active zone. The active zone is the 3D space within which the shadow hand can move its end effector. This area is determined by the arm's mechanical structure, including the number, type, and range of motion of its joints. Since the active zone is relatively confined and finite it is better to use it to find deadzones as the deadzone would consist of all of 3D space minus the small zone where the robot can articulate.

We tried several variables to get a sense of its physical attributes. We tried Cartesian position, velocity, potential energy, kinetic energy, contact forces and number of forces. The potential energy, kinetic energy and contact forces were all blank for

each iteration.

We figured out that because we were using the `data.ctrl[id of link]`, it only changed the angle of rotation on each link, meaning that no outside force was acting on it. This contradicted the simulation render which showed contact forces, but were able to gain insight a different way.

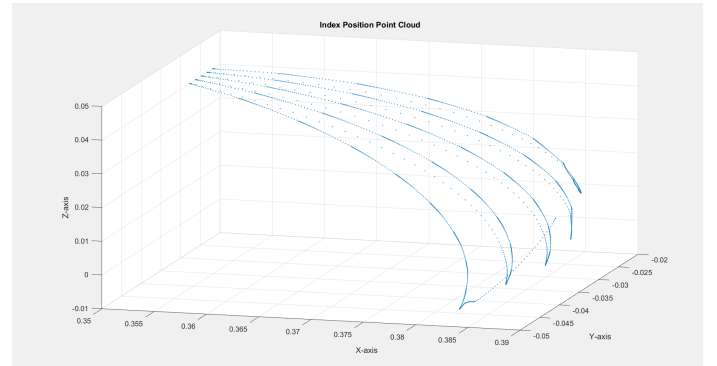


Figure 1: Index Finger Position Point Cloud

Velocity

Because we were only changing position in equidistant increments, the velocity should have been a constant value throughout the simulation. However, the finger that was articulating would either not be able to move when in contact with another surface or sometimes get caught on a different finger. When this happened, the articulating finger would quickly correct itself to match up with the proper rotational value dictated by the control variable.

Using the edges of the active zone we can map out the borders of the deadzone for each finger. When a velocity on this map that does not match up to the normal small velocity we can say its on the edge of the deadzone.

We can also see the effects of these deadzones by seeing where in the finger doesn't go in a linear curve but ricochets back and forth slightly.

Contacts

We also found an attribute of the data that said how many contacts are being recognized within the system. This was even more insightful as only contact with a surface would produce a different color which would even more physically coincide with

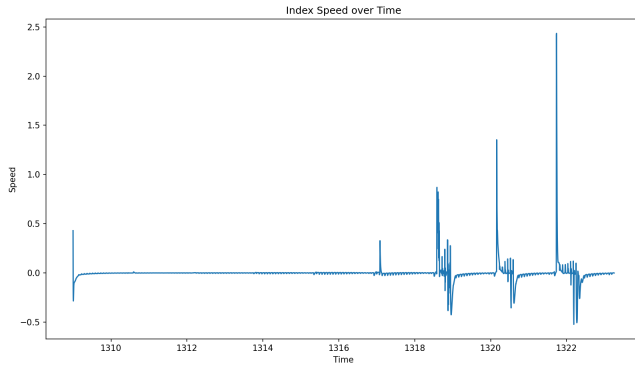


Figure 2: Index Finger Speed Over Time

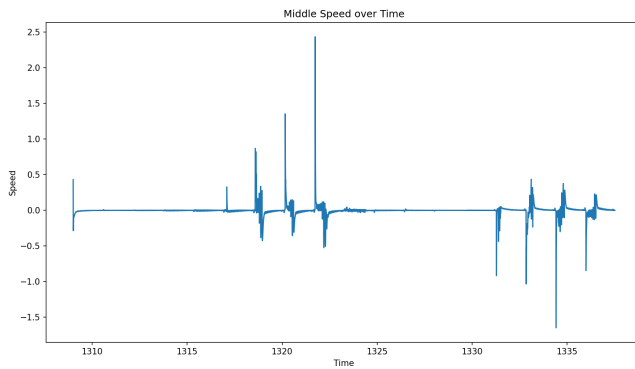


Figure 3: Middle Finger Speed Over Time

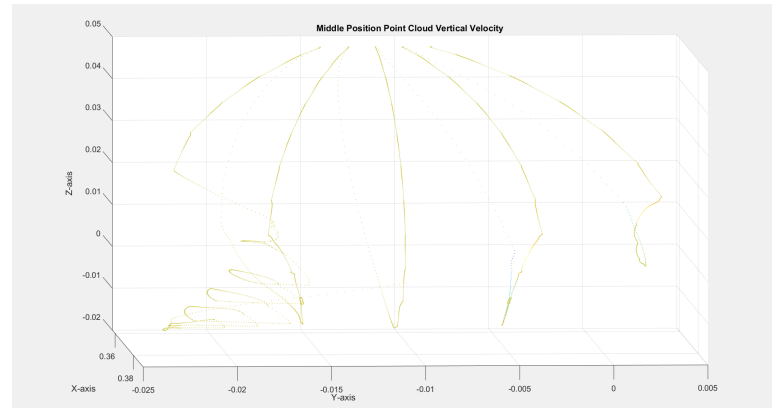


Figure 5: Middle Finger Position Colored with Velocity

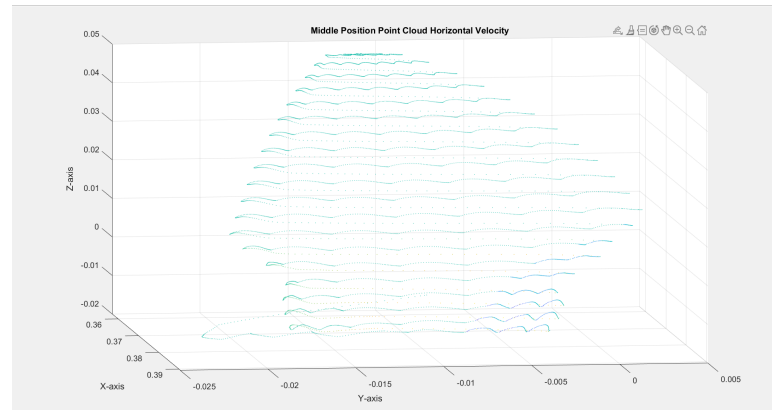


Figure 6: Middle Finger Position

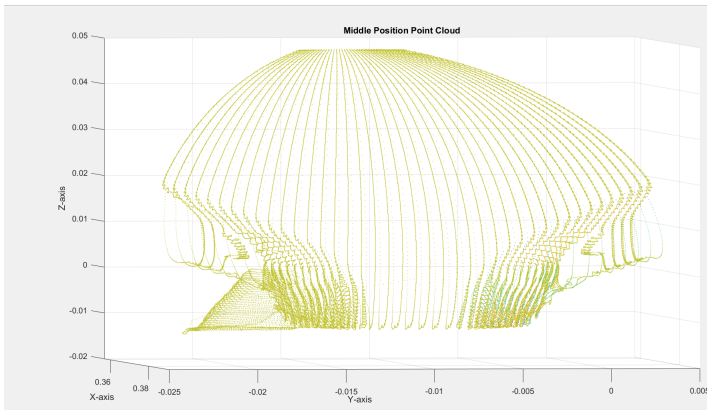


Figure 4: Middle Finger Position Colored with Velocity

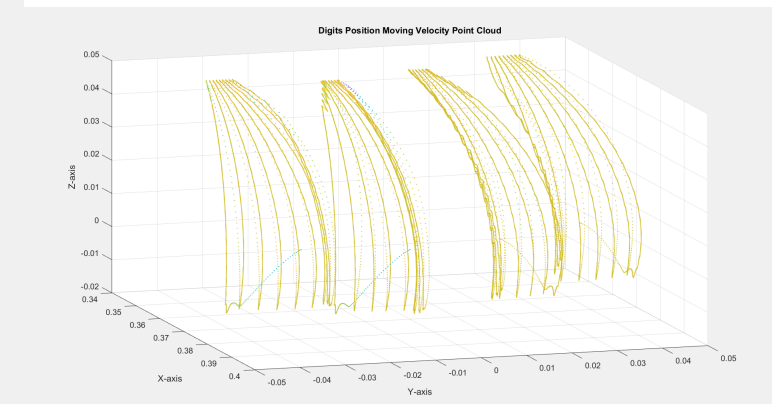


Figure 7: All Digits

a deadzone border. Mapping out all the positions of each fingers we could see how each finger related to one another in physical space. More color weights indicates more contacts which imply that the finger's normal active zone is being blocked by another object.

This colored graph was slightly off color since it was detecting contact with the ball object in the scene as well.

On these graphs, a color change is only clearly shown on a finger's border of movement. On the index finger, its active zone is cut off by the middle fingers active zone.

The major epiphany we had was that the deadzones for each

finger are ever changing and physically dependent on the articulation of each adjacent finger. When a finger reaches a deadzone or gets stuck, it still tries to reach its control destination. This can be seen from the areas where the path the end effector takes are not on a linear/ repeating pattern along the curving path each of the digits saddle shaped active zone.

Next steps

The next steps of this project would be to take the accounted deadzone and run it through a precompensator to shift the deadzone. Using the dynamic and modeling and our simulation environment for the dextrous shadow hand, we could hypo-

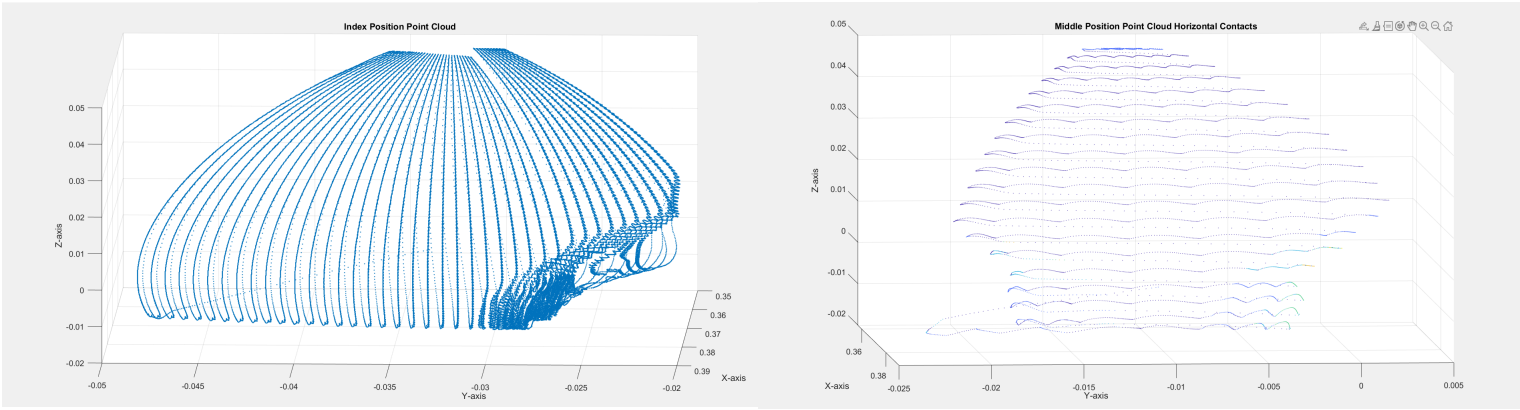


Figure 8: Index Position Point Cloud

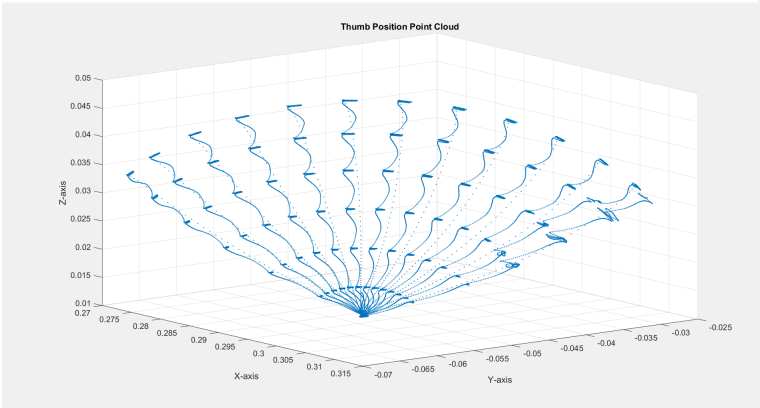


Figure 9: Thumb Position

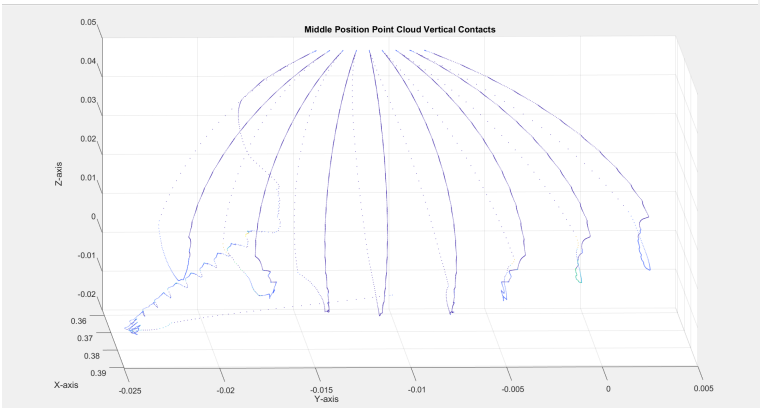


Figure 10: Thumb Position

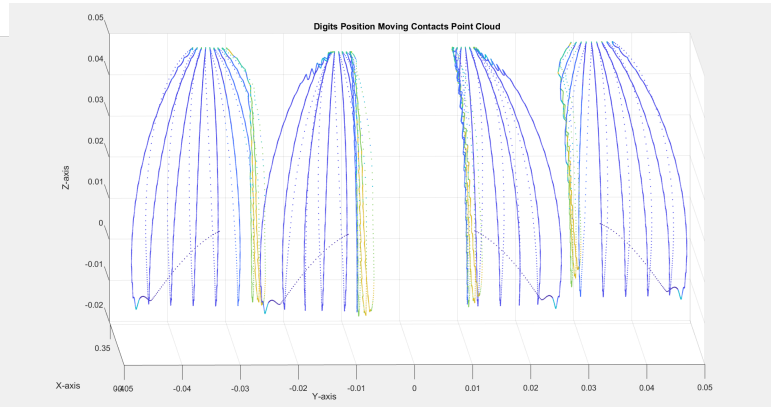


Figure 11: All Digits Moving

thetically create a transfer function with a feedback loop to avoid the dead zone [7, 5]. In control research, non-smooth non-linearities in actuators such as dead zones and singularities can result in imprecision in motion. By accounting for this, we can pre-compensate the hand to alleviate motions through dead zones better. This has long been a hurdle in control research, with most commonly implemented systems ignoring dead zones due to their time-variant nature. This approach intends to smooth the non-linearities to improve control performance instead of just trying to offset the deadzone.

To start off, take the following nonlinear dynamic model of an n rigid-link robot system with direct-drive joints is given in the following Lagrange form as seen in FIG ??.

Here we see q as the joint position, $\dot{q}$ is joint velocity, $\ddot{q}$ is acceleration. M is the moment of inertia matrix and a positive definite symmetric matrix. C is the centripetal Coriolis matrix. G is the gravity vector. $T.L$ is the external disturbance. $T.F$ is the friction which will be ignored. When we ignore links affecting each other and take each individually we see FIG 15.

By taking the accounted deadzone and running it through a precompensator we can better define the deadzone width. This will help us define the block diagonal matrix of the deadzone, $D = [d_+, d_-]^T$, and offset the "deleterious" effects as seen in FIG 16.

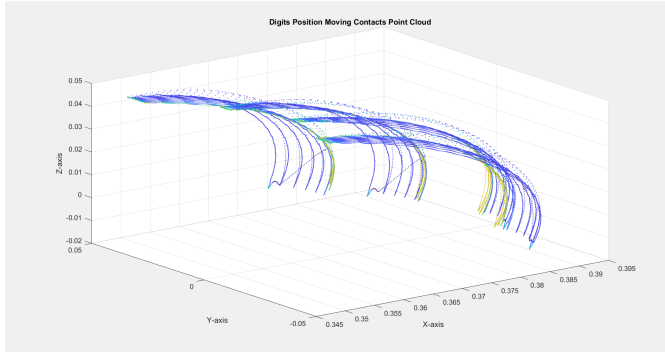


Figure 12: All Digits Moving

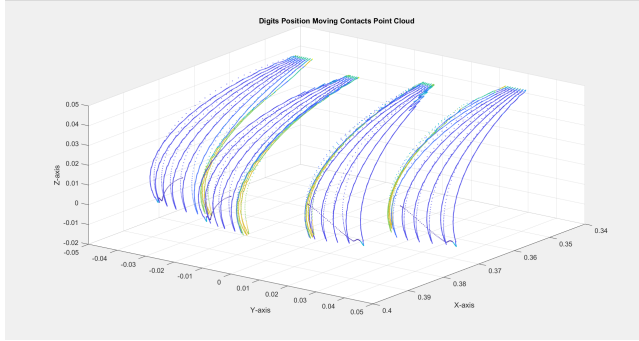


Figure 13: All Digits Moving

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{T}_f(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{T}_L = \boldsymbol{\tau}$$

Figure 14: Lagrange Form for N Rigid-Link System

$$\boldsymbol{\tau} = [\tau_1 \quad \tau_2 \quad \cdots \quad \tau_n]^T = \mathbf{D}_D(\boldsymbol{\tau}_d) .$$

Figure 15: Predicted deadzone width in relation to torque given independent links.

$$\tau_i = D_d(\tau_{di}) = \begin{cases} \tau_{di} - d_{i+} & \text{for } d_{i+} \leq \tau_{di} \\ 0 & \text{for } d_{i-} \leq \tau_{di} \leq d_{i+} \\ \tau_{di} + d_{i-} & \text{for } \tau_{di} \leq -d_{i-} \end{cases}$$

Figure 16: Precompensator to help offset the deadzone.

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